Homework 8: Price an option from history alone

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Data: SPY unadjusted daily closes from yfinance, 2006-11-09 to 2026-11-09 (the Nov 10 session was still trading when this ran, so it is left out). Unadjusted closes are the right series because an SPY option settles on the price and its holder does not receive dividends. The spot of $760.88 is the figure the assignment gives; SPY closed at $752.69 on Nov 9. check_hw08.py passes 10 of 10.

1. The 21-day return distribution

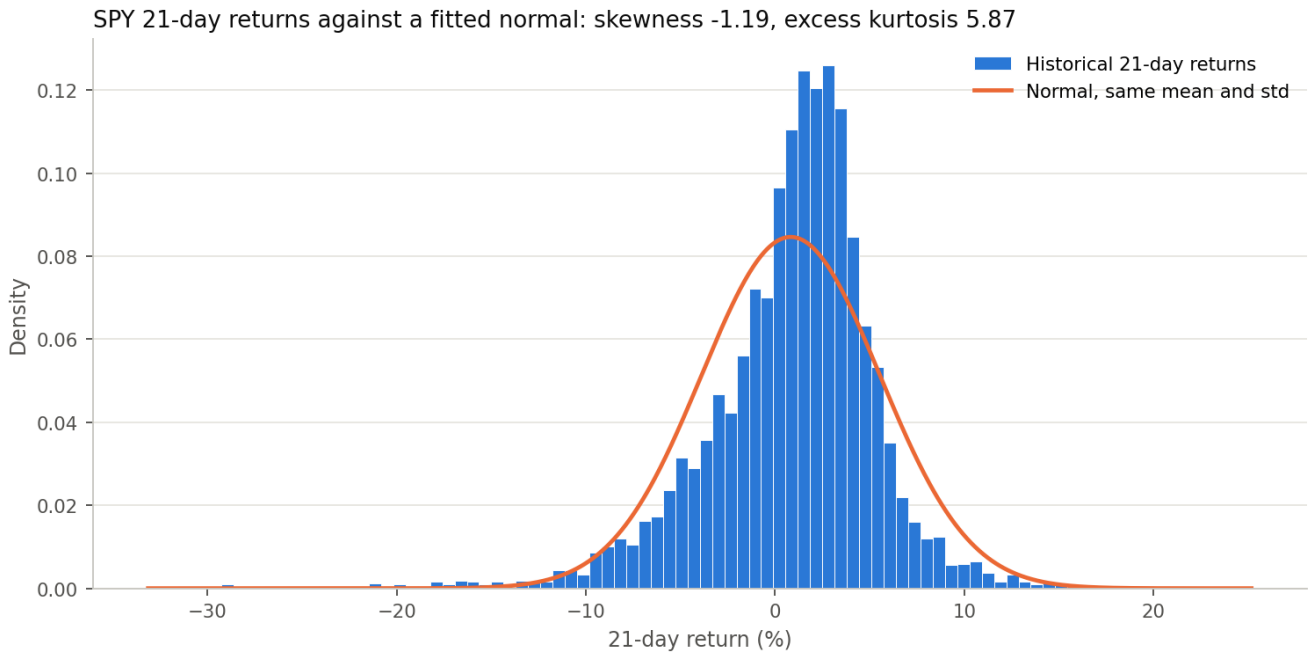
5,030 trading days give 5,009 overlapping 21-day windows. The mean is +0.83% and the standard deviation 4.72%, with a 1st percentile of -14.81% and a 99th percentile of +11.24%.

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Q1  SPY daily closes 2006-11-09 to 2026-11-09: 5030 trading days (unadjusted closes, the price an option actually
    settles on)
    N = 21: 5009 windows
    mean +0.8282%   std 4.7152%   1st percentile -14.81%   99th percentile +11.24%
    worst window -33.14% (starting 2020-02-21), best window +25.19% (starting 2020-03-23)
```

2. Histogram against a fitted normal

Excess kurtosis is 5.87, well above zero, so the distribution has fatter tails and a taller, narrower peak than a normal with the same mean and standard deviation. Skewness of -1.19 puts the long tail on the left: the 1st percentile sits 3.32 standard deviations below the mean where a normal would put it 2.33 below, while the 99th is only 2.21 above. Daily returns have excess kurtosis of 14.55, as the check predicts. Summing to 21 days pulls that down but nowhere near zero.

```
Q2  21-day returns: skewness -1.188, excess kurtosis 5.873  (above zero, so fatter tails than normal)
    daily returns:  skewness -0.008, excess kurtosis 14.547
    the 1st percentile sits -3.32 std from the mean and the 99th +2.21 (a normal gives -2.33 and +2.33)
```



3. The historical pricer

The 21-day call struck 3% above spot ($783.71) prices at $5.36 and the matching put at $21.89. **These prices are not discounted.** The omission overstates both by the discount factor, 0.99664 at the 4.035% T-bill rate, which is 0.34%: $0.02 on the call and $0.07 on the put. It is small because the horizon is short.

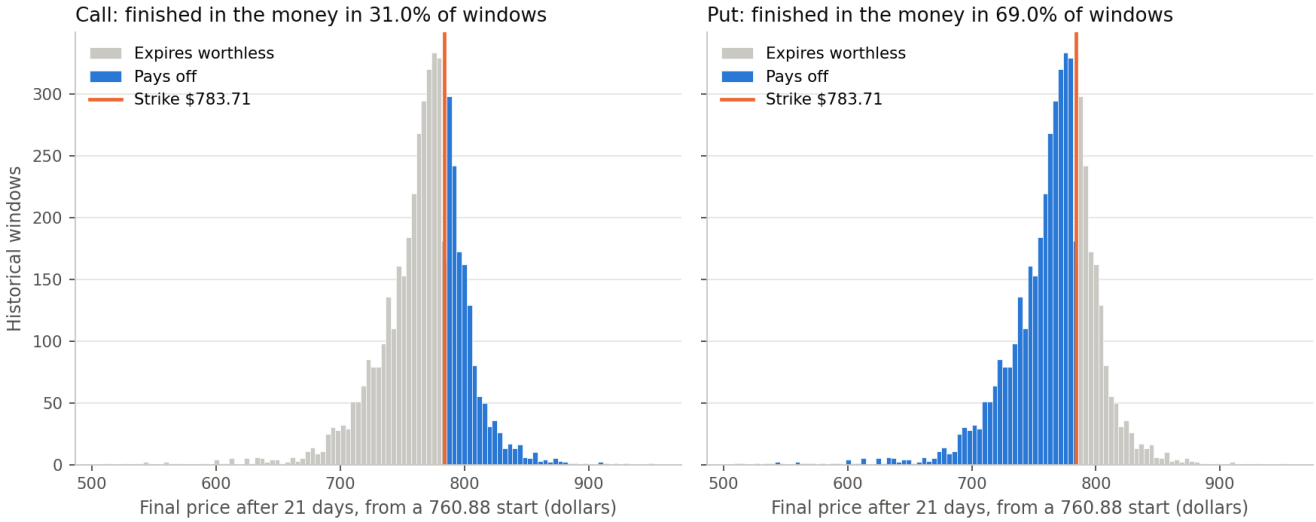
The bigger issue is that the returns keep their real-world drift. The call minus the put is -$16.52 rather than the -$22.83 of spot minus strike, and the +$6.30 gap is exactly spot times the +0.83% mean return. Using dividend-adjusted closes instead would add $0.37 to the call and take $0.80 off the put, the same effect in a smaller size.

```
Q3  spot 760.88, strike 783.71 (3% above spot), 21 days, undiscounted
    historical call 5.3618
    historical put  21.8863
    call - put = -16.52;  spot - strike = -22.83;  spot x mean return = +6.30;  those two sum to -16.52
    discounting at 4.035% for 21 trading days multiplies by 0.99664: call 5.3438 (-0.0180), put 21.8129 (-0.0735)
    with dividend-adjusted closes instead: call 5.7344, put 21.0856
```

4. Distribution of final prices

The call would have finished in the money in 31.02% of windows and the put in 68.98%.

```
Q4  fraction of the 5009 windows finishing in the money at strike 783.71:
    call 31.02%   put 68.98%
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5. Black-Scholes on the same contracts

Inputs are the 14.80% realised volatility of the last 60 daily log returns, the 4.035% 13-week T-bill as in Homework 6, and T = 21/252. The 60-day figure includes the -4.41% day on Nov 5, and it would be 11.62% without that one day.

The call differs by only -1.4%, but that agreement is partly luck: drift pushes the historical call up while the narrow peak of the distribution pulls it down. The put differs by -14.6%, and nearly all of that is parity. Of its -$3.75 gap, -$3.67 is drift net of interest (spot times the mean return, less strike times one minus the discount factor), which leaves the same -$0.08 as the call.

```
Q5  Black-Scholes inputs: realised vol over the last 60 days 14.80%, rate 4.035% (13-week T-bill, 2026-11-09), T =
    21/252 years
    contract    historical  Black-Scholes   hist - BS    % diff
    call           5.3618         5.4374     -0.0756     -1.4%
    put           21.8863        25.6330     -3.7467    -14.6%
    % diff is (historical - Black-Scholes) / Black-Scholes
    call - put: historical -16.52, Black-Scholes -20.20, no-arbitrage value spot - strike x discount -20.20
    same contracts at r = 0: Black-Scholes call 4.7790, put 27.6054
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6. Across strikes

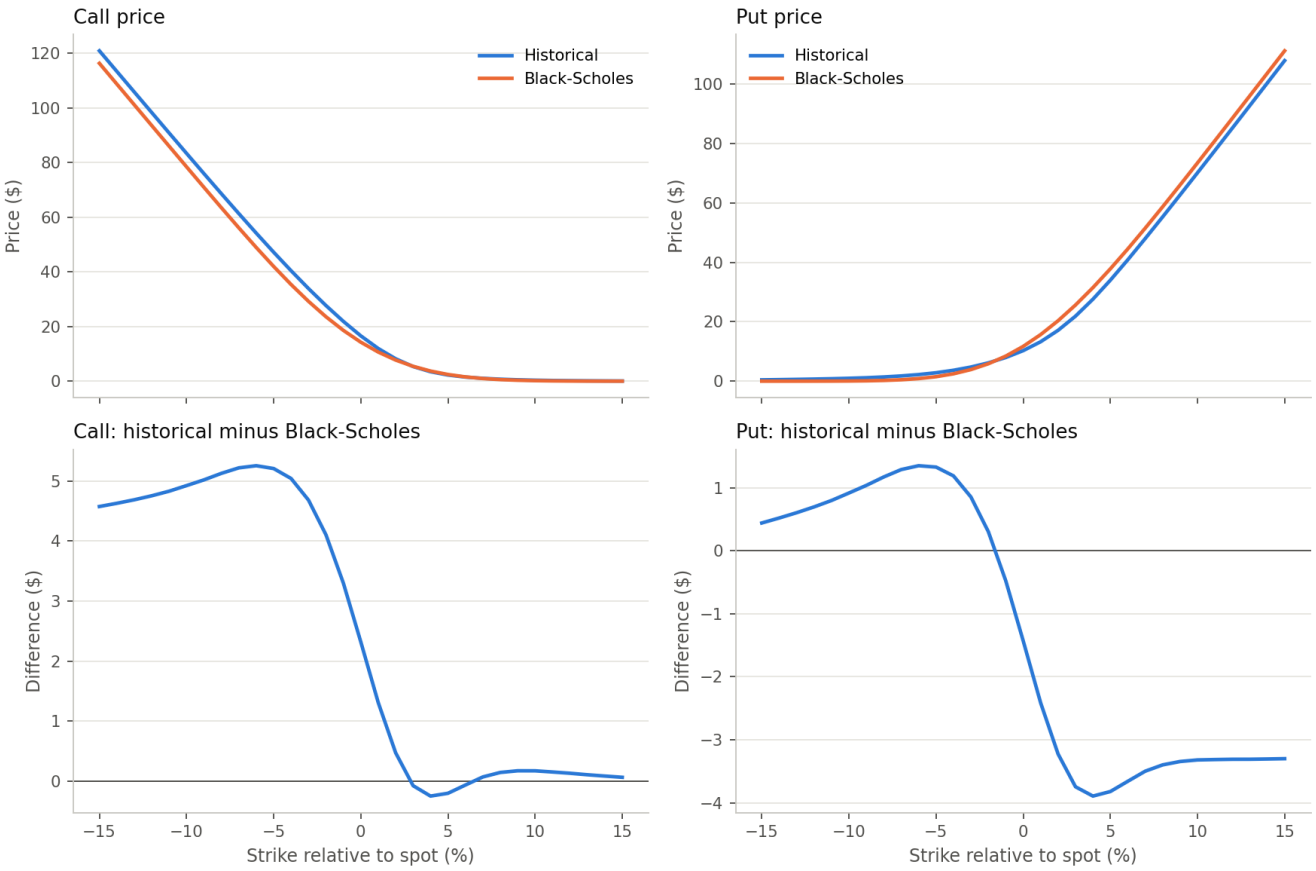
In dollars, the methods disagree most on the call struck at $715.23 (94% of spot), by +$5.25. For puts the largest gap is -$3.89 at $791.32 (104% of spot). Those are in-the-money contracts and the gap is mostly drift. At every strike, parity forces the call gap minus the put gap to equal that same drift-less-interest term, about $4.13 at the 85% strike and $3.36 at 115%.

In proportion, the disagreement is largest on the deep out-of-the-money puts, which is the effect the assignment points to. At 85% of spot the historical put is $0.443 against $0.000358 for Black-Scholes, 1,240 to 1 (log-scale plot below). History says SPY fell more than 15% over 21 days in 0.978% of windows. A lognormal at 14.80% volatility says 0.0056%, about 175 times less, because a single volatility makes that a 3.86 standard deviation event. Crashes do not fit that distribution.

It is the same reason the implied volatility curve in Homework 6 sloped up to the left. Solving for the volatility that reproduces each historical out-of-the-money price gives 28.3% at the 85% strike, 22.8% at 90%, 17.9% at 95% and 13.2% at the money. The Homework 6 market quotes were 32.1%, 25.4%, 20.1% and 14.5% at about the same strikes (a 32-day expiry, so a loose comparison). The market charges for the left tail that one flat volatility cannot produce. The right side does not match: the market flattened near 12%, while history rises to 18.4% at 115% because its largest up-moves are the 2009 and 2020 rebounds.

```
Q6  strikes from 15% below spot to 15% above, in 1% steps
      strike  % spot |  call hist   call BS     diff |  put hist    put BS     diff      % diff
      646.75     85% |    120.877   116.303   +4.574 |     0.443     0.000   +0.443   +123,852%
      654.36     86% |    113.348   108.721   +4.627 |     0.523     0.001   +0.522    +45,958%
      661.97     87% |    105.826   101.140   +4.686 |     0.609     0.003   +0.606    +18,202%
      669.57     88% |     98.315    93.562   +4.752 |     0.707     0.009   +0.698     +7,720%
      677.18     89% |     90.822    85.993   +4.829 |     0.823     0.023   +0.800     +3,506%
      684.79     90% |     83.362    78.440   +4.922 |     0.973     0.054   +0.919     +1,712%
      692.40     91% |     75.939    70.922   +5.018 |     1.158     0.118   +1.040       +881%
      700.01     92% |     68.590    63.464   +5.126 |     1.418     0.244   +1.174       +482%
      707.62     93% |     61.328    56.110   +5.218 |     1.764     0.473   +1.291       +273%
      715.23     94% |     54.174    48.921   +5.253 |     2.219     0.867   +1.352       +156%
      722.84     95% |     47.181    41.975   +5.206 |     2.835     1.505   +1.331        +88%
      730.44     96% |     40.409    35.368   +5.040 |     3.672     2.481   +1.191        +48%
      738.05     97% |     33.883    29.203   +4.680 |     4.755     3.899   +0.856        +22%
      745.66     98% |     27.682    23.579   +4.103 |     6.162     5.858   +0.304         +5%
      753.27     99% |     21.877    18.580   +3.297 |     7.966     8.443   -0.476         -6%
      760.88    100% |     16.582    14.264   +2.318 |    10.280    11.709   -1.430        -12%
      768.49    101% |     11.959    10.651   +1.309 |    13.266    15.680   -2.414        -15%
      776.10    102% |      8.197     7.726   +0.471 |    17.113    20.338   -3.225        -16%
      783.71    103% |      5.362     5.437   -0.076 |    21.886    25.633   -3.747        -15%
      791.32    104% |      3.461     3.710   -0.249 |    27.595    31.489   -3.894        -12%
      798.92    105% |      2.250     2.452   -0.202 |    33.993    37.815   -3.822        -10%
      806.53    106% |      1.505     1.570   -0.065 |    40.856    44.515   -3.659         -8%
      814.14    107% |      1.043     0.973   +0.070 |    48.003    51.502   -3.499         -7%
      821.75    108% |      0.728     0.584   +0.145 |    55.297    58.695   -3.399         -6%
      829.36    109% |      0.511     0.339   +0.173 |    62.689    66.034   -3.345         -5%
      836.97    110% |      0.363     0.190   +0.172 |    70.149    73.469   -3.320         -5%
      844.58    111% |      0.256     0.104   +0.152 |    77.651    80.965   -3.314         -4%
      852.19    112% |      0.186     0.055   +0.131 |    85.190    88.500   -3.310         -4%
      859.79    113% |      0.134     0.028   +0.106 |    92.747    96.056   -3.309         -3%
      867.40    114% |      0.099     0.014   +0.085 |   100.320   103.625   -3.305         -3%
      875.01    115% |      0.071     0.007   +0.064 |   107.901   111.201   -3.300         -3%
    call: largest dollar disagreement at strike 715.23 (94% of spot): historical 54.174, Black-Scholes 48.921,
        difference +5.253
    put: largest dollar disagreement at strike 791.32 (104% of spot): historical 27.595, Black-Scholes 31.489,
        difference -3.894
    overall, in dollars: the call struck at 715.23 (94% of spot), by +5.253
    deep out-of-the-money puts, in proportion: at strike 646.75 (85% of spot) historical 0.443 against Black-
        Scholes 0.000358, a ratio of 1,240 to 1

    Black-Scholes vol that reproduces each historical out-of-the-money price (60-day realised is 14.80%):
       85% of spot (put ): 28.26%
       90% of spot (put ): 22.79%
       95% of spot (put ): 17.87%
      100% of spot (put ): 13.16%
      105% of spot (call): 14.40%
      110% of spot (call): 16.32%
      115% of spot (call): 18.42%
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7. Written

The two methods agree closely on calls above about 102% of spot and on puts just below spot, within roughly $0.50. They disagree in two different ways. Away from the money on the in-the-money side, the $3 to $5 gaps come from drift: history keeps the +0.83% monthly mean where Black-Scholes grows the stock at the risk-free rate, and that is a defect of the historical method rather than information about the tails. On deep out-of-the-money puts, the gap is a genuine difference in shape, since history contains crashes and the lognormal effectively does not. Some of the gap is level rather than shape, because history's standard deviation annualises to 16.3% against the 14.8% recent

volatility, so the comparison also sets a long-run average against a fairly calm recent stretch.

For a contract 20% out of the money I would trust history over this Black-Scholes, as a rough estimate and not as a price. Black-Scholes prices the 80% put at $0.0000003, treating a 20% monthly fall as essentially impossible, though it happened in both 2008 and 2020. History's $0.19 at least has the right order of magnitude, and the market agrees in direction: in Homework 6 the strike at 79.7% of spot was quoted at 39% volatility. I would not lean on either for the 20% out-of-the-money call, since the three windows behind history's $0.014 are all rebounds straight after crashes. One specific weakness of the historical method that Black-Scholes does not have is that it is not arbitrage-free. Using real-world returns leaves the historical call minus put at -$16.52, while no-arbitrage requires spot minus discounted strike, -$20.20. That $3.67 violation of put-call parity could be traded against, and Black-Scholes satisfies parity by construction.

The overlapping windows make the sample look far larger than it is. Consecutive windows share 20 of 21 days, so 5,009 windows hold about as much independent information as 240, and standard errors computed as if they were independent are too small by about the square root of 21, a factor of 4.6. In the tail that matters most, the damage is severe. The 22 windows below -20% are about one independent window, drawn from just two episodes. Non-overlapping samples contain 0, 1 or 2 such events depending only on which day the count starts, and the 80% put then prices anywhere from $0.00 to $0.62. The standard error on that 0.44% probability is 0.093% if the windows are treated as independent but 0.427% using the effective count, about as large as the estimate itself. The $0.19 really means somewhere between nothing and several times that, and even 240 is generous because crises cluster.

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Q7  how much independent evidence sits in the tails (21-day windows, strikes as a % move from spot)
    5009 overlapping windows contain about 240 non-overlapping ones
    below -15%: 49 windows (0.98%), 4 distinct episodes, about 2.3 independent windows (non-overlapping counts run
        1 to 4 depending on the start day)
        2008-09-05 to 2008-12-04:  21 windows, extreme -30.5%
        2009-01-28 to 2009-03-11:   8 windows, extreme -19.5%
        2011-07-08 to 2011-08-22:   3 windows, extreme -16.5%
        2020-02-06 to 2020-04-03:  17 windows, extreme -33.1%
    below -20%: 22 windows (0.44%), 2 distinct episodes, about 1.0 independent windows (non-overlapping counts run
        0 to 2 depending on the start day)
        2008-09-08 to 2008-10-27:  12 windows, extreme -30.5%
        2020-02-11 to 2020-03-25:  10 windows, extreme -33.1%
    above +20%: 3 windows (0.06%), 2 distinct episodes, about 0.1 independent windows (non-overlapping counts run
        0 to 1 depending on the start day)
        2009-03-05 to 2009-04-06:   2 windows, extreme +22.5%
        2020-03-23 to 2020-04-22:   1 windows, extreme +25.2%
    chance of finishing more than 15% down: history 0.978%, Black-Scholes at 14.80% vol 0.0056% (-3.86 std)
    standard error on the 0.44% chance of a 20% fall: 0.093% if the windows were independent, 0.427% using the
        non-overlapping count
    prices 20% out of the money:
      put at 80% of spot: historical 0.1851, Black-Scholes 2.95e-07; non-overlapping windows alone give 0.0000 to
          0.6182 depending on the start day
      call at 120% of spot: historical 0.0136, Black-Scholes 1.08e-04; non-overlapping windows alone give 0.0000
          to 0.1651 depending on the start day
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Check yourself

```
trading days 5030 (about 5,030)
21-day windows 5009 (about 5,010)
daily excess kurtosis 14.55 (well above 3)
call - put -16.52 against spot - strike -22.83; the +6.30 gap is exactly spot x the mean 21-day return
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